SOUTHERN ADVENTIST UNIVERSITY
Physics and Engineering Department

ENGR 330: Thermodynamics

Project – Fall 2026

Background:

This project is an opportunity for you to explore applications of thermodynamics. Principles of thermodynamics are found widely throughout engineering and science and act as critical tools in the engineer's and scientist's toolbox. As a major project constituting 10% of the class grade, this project is expected to take at least 10 hours of work. Spread over the last 5 weeks of the semester, this comes to 2 hours per week. Students are free to utilize existing lab equipment, propose a new experiment, perform numerical analysis of an existing or proposed system, run simulations for a thermodynamics-related system or problem, or propose something entirely out of the box.

Proposal Requirements:

A 1 – 2 page, single-spaced, no additional line spacing, 10-point, Times New Roman (or Georgia) font, 1-inch margin document with three sections

> **Background:** This section should provide some introductory context on the problem or the area of research that you're choosing. This should include why you're interested in the problem or area of research, why you believe the problem or area of research is relevant to engineering or science, and any initial information that you can find regarding what has been done about the problem in the past. This section should have at least 3 references with citations in IEEE format.
> **Problem Statement and Deliverables:** This should be a clear definition of the problem and any necessary details or constraints. It should also clearly state the deliverables and what you will do in this project.
> **Methods/Approach:** This section is a statement of how you will approach the problem or area of research. This should include information about how you will use lab equipment, construct an experiment or an experimental apparatus, perform simulations or analysis, and how you will analyze data. This section should also motivate further developments (that is, you can state open, unknown questions or areas that you will investigate further in order to achieve the deliverables stated in the previous section).
> **References:** A list of citations in IEEE format at the end of the proposal.

Project Proposal Due Date: October 26, 2026

The project proposal is due on eClass and in-person, Monday morning, at the start of class, October 26, 2026.

Project Requirements:

All projects:

- No AI may be used during the execution of any phase of this project without prior consent from the professor. If the professor deems that your proposed usage of AI is acceptable, then you will

need to rigorously document all AI usage. This means recording the date, time, model used (e.g., Anthropic Sonnet 5.5, OpenAI GPT-5.5), the prompt used (full text of the prompt), and the AI model output.

- No off-the-shelf solutions may be used during the execution of any phase of this project. That means, you cannot use already existing analyses or pre-existing solution paths.

- This is an individual project. If you want to work with a classmate, the project that you design, and/or construct, and and/or test, and/or research will have to be at least 2x as complex and challenging as the baseline project expected for students and you will have to obtain instructor permission. If you work in larger teams, the complexity will scale with the size of the team.

- You must use thermodynamic analyses for the project. This means that the project should revolve around thermodynamics in some aspect and should be able to be explored through either macroscopic or microscopic thermodynamics.

Examples:

- **Mechanical Lab Experiments – Steam Power Plant Analysis:**

  Run the steam power plant and collect data. Calculate the overall efficiency of the system and identify efficiencies and losses associated with each component in the power plant (boiler, steam lines, steam turbine, condenser). Approximate the rough scaling of efficiency with respect to the size of the system, the operating temperatures of the system, and the scale of the steam lines. Propose improvements to the steam power plant and analyze the theoretical improvement these would provide to the system.

- **Mechanical Lab Experiments – Refrigeration Cycle Analysis:**

  Run the refrigeration test stand and collect data. Calculate the overall coefficient of performance of the system and identify efficiencies and losses associated with each component. Derive correlations for coefficient of performance versus temperature differences from multiple operating conditions. Propose improvements to the refrigeration cycle test stand and analyze the theoretical improvement these would provide to the system.

- **Mechanical Lab Experiments – Heat Exchanger Analysis:**

  Run the heat exchanger test stand and collect data from both parallel and counterflow . Calculate the internal energy transfer from the hot side to the cold side of the exchanger and an associated effectiveness and performance with respect to the maximum possible heat transfer.

- **Dehumidifier Analysis:**

  Analyze the performance characteristics of a consumer-grade dehumidifier. Calculate the energy transfers, efficiencies, and overall coefficient of performance. Obtain a small consumer-grade dehumidifier, tear it apart, and document the overall design, including engineering considerations and characteristics.

- **HVAC System Performance Analysis:**

Similar to dehumidifier analysis except on an HVAC system (either in a home or in a vehicle). The main difference is that a teardown won't be as easily performed (maybe not possible at all, except for the outdoor unit)

- **Air Compressor Analysis:**

  Similar to the previous two except, most likely, no teardown (exceptions could be found).

- **Vortex Tube Analysis:**

- **Campus Building Energy and Second-Law Audit:**

  Similar to HVAC system performance analysis except that it would be at the whole-building scale, wouldn't require teardown, would be working with Plant Services. You could possibly convince Dr. Howard to work with you as well.

- **Alternative Working Fluid Analysis (ORC, low GWP refrigerants):**

  System-level design of HVAC or power systems using organic Rankine cycles or other alternative working fluids.

- **Thermodynamic Cycle Analysis for Cycles Not Covered in Class:**

- **Simulation and Analysis of Rocket Nozzle:**

- **Transient Simulations of Piston-Cylinder with Heat Transfer:**

- **Analyses Utilizing Alternative Equations of State (van der Waals, Redlich-Kwong):**

- **Whole-Plant Energy Audit Model for Rankine, Brayton, or Combined Cycle Plant:**

- **Designs of Thermodynamic Systems to Meet Specified Need (plant, refrig, etc.)**

- **Waste-Heat Recovery Analysis:**

- **Fusion-Relevant Power Conversion Cycle Model (Supercritical CO_2, etc.)**

- **Heat Pump vs. Electric vs. Natural Gas Heating Across US Climate Zones:**

  Cost and thermal performance analysis on common heating systems in the US.

- **Central AC vs. Evaporative Cooler Across US Climate Zones:**

  Cost and thermal performance analysis on common cooling systems in the US.

Project Report:

A single-spaced, no additional line spacing, 10-point, Times New Roman (or Georgia) font, 1-inch margin document with:

Background: This section should provide enough introductory context on the problem or the area of research that you're choosing to orient the reader to the project. This should include why the problem or area of research is relevant to engineering or science, and information regarding what has been done about the problem in the past. This section should have at least 5 references with citations in IEEE format. This section should end with a clearly defined problem statement and any necessary constraints or requirements.
Methodology: This section is a statement of how you approached the problem or area of research. This should include research approaches, design methods, tools and materials utilized, procedures, and how the results or data was analyzed or evaluated.
Results and Discussion: This section should discuss the results and developments during the course of the project and the implications of the results and developments.
Conclusion and Future Work: The paper should end with conclusions drawn from the project and future work that can be pursued as a result of this project.
References: A list of citations in IEEE format at the end of the proposal.

Grading:

Grading will be based on the following items:

- Quality of the report (structure, formatting, flow, etc.): this should be viewed as a formal lab report that you would be willing to present as an example of your work during a future interview
- Quality of the research question and the research analysis
- Extra credit points can be earned for (a) on-topic, engineering creativity, (b) particularly unique projects, and (c) very high quality reports.

Resources:

- The course textbook
- Online examples for getting ideas for projects, not ideas of replication
- Examples provided in class
- Ideas from professors
- Ideas from lab experiment manuals